Problem 1.9

(a) Take A to be a trifling set and take an open interval I such that $A \subset I$. Then take a finite number of intervals $I_1, \ldots, I_n$ such that

$$I \subset \bigcup_{i=1}^{n} I_n$$

and $\sum_{i=1}^{n} |I_n| < |I|/2$. Further $I \setminus \bigcup_{i=1}^{n} I_i$ is a union of intervals, which yields the claim.

(b)
- $(0,1] - B$ is nowhere dense: Take any interval I such that $(0,1] - B \subset I$. Let $I = (a, b]$, then take $\varepsilon > 0$ such that $(a + \varepsilon, b - \varepsilon]$ is non-empty. Then there exist infinitely many rationals in $(a+\varepsilon, b-\varepsilon]$ and take one within the enumeration such that $r_n \in (a+\varepsilon, b-\varepsilon]$ and $2^n < \varepsilon$. This means $(r_n - 2^{-n-2}, r_n + 2^{-n-2}) \subset I$ by construction and $(r_n - 2^{-n-2}, r_n + 2^{-n-2})$ does not meet $(0,1] - B$.
- $(0,1] - B$ is not negligible: If it were negligible, then for $\varepsilon > 0$ there would exist intervals I_n of $(0,1]$, for $n \in \mathbb{N}$ such that

  $$(0,1] - B \subset \bigcup_{n} I_n$$

  and

  $$\sum_{n} |I_n| < \varepsilon\,.$$

  This implies that (we take complements wrt. $(0,1]$)

  $$\bigcap_{n} I_n^c \subset (0,1] \cap B\,,$$

  which immediately implies

  $$1 - \varepsilon \leq |I \cap B|$$

  for all $\varepsilon > 0$, however a simple union bound on $I \cap B$ (to be formally precise at this stage refer to Theorem 1.3) yields

  $$|I \cap B| \leq \sum_{n \geq 1} 2^{-n-1} = \frac{1}{2}\,,$$

  which would yield a contradiction.

(c) This immediately follows by every open covering of a compact set having a finite sub-covering. In the definition of a negligible set, expand the intervals by $+\varepsilon 2^{-n}$ to relate the definition to open coverings.